极限存在准则与重要极限

夹逼定理

$$\text{若}a_n \le b_n \le c_n$$
$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = A \Rightarrow \lim_{n\to\infty} b_n = A$$

$$\text{证}: \forall\epsilon > 0, \exists N_1 > 0, \text{当}n > N_1\text{时},$$
$$|a_n - A| < \epsilon \Leftrightarrow A - \epsilon < a_n < A + \epsilon(*)$$
$$\exists N_2 > 0, \text{当}n > N_2\text{时},$$
$$|c_n - A| < \epsilon \Leftrightarrow A - \epsilon < c_n < A + \epsilon(**)$$
$$\text{取}N = \max\{N_1, N_2\}, \text{当}n > N\text{时}, (*)(**)\text{成立},$$
$$\text{当}n > N\text{时}$$
$$A - \epsilon < a_n \le b_n \le c_n < A + \epsilon$$
$$\Rightarrow A - \epsilon < b_n < A + \epsilon \Leftrightarrow |b_n - A| < \epsilon$$
$$\therefore \lim_{n\to\infty} b_n = A$$

$$\text{若}f(x) \le g(x) \le h(x)$$
$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = A$$
$$\Rightarrow \lim_{x\to a} g(x) = A$$

$$\text{证}: \forall\epsilon > 0$$
$$\exists\delta_1 > 0, \text{当}0 < |x - a| < \delta_1\text{时},$$
$$|f(x) - A| < \epsilon \Leftrightarrow A - \epsilon < f(x) < A + \epsilon(*)$$
$$\exists\delta_2 > 0, \text{当}0 < |x - a| < \delta_2\text{时},$$
$$|h(x) - A| < \epsilon \Leftrightarrow A - \epsilon < h(x) < A + \epsilon(**)$$
$$\text{取}\delta = \min\{\delta_1, \delta_2\}, \text{当}0 < |x - a| < \delta\text{时}, (*)(**)\text{皆对}$$
$$A - \epsilon < f(x) \le g(x) \le h(x) < A + \epsilon$$
$$\Rightarrow A - \epsilon < g(x) < A + \epsilon \Leftrightarrow |g(x) - A| < \epsilon$$
$$\lim_{x\to a} g(x) = A$$

$$\text{求极限} \lim_{n\to\infty} \sqrt[n]{2^n + 3^n}.$$

$$3^n \le 2^n + 3^n \le 2 * 3^n \Rightarrow 3 \le \sqrt[n]{2^n + 3^n} \le 2^{\frac{1}{n}} * 3$$
$$\because \lim_{n\to\infty} 2^{\frac{1}{n}} = 2^0 = 1, \therefore \lim_{n\to\infty} 3 = 3 = \lim_{n\to\infty} 2^{\frac{1}{n}} * 3 = 3$$
$$\therefore \text{原式} = 3$$

$$a > 0, b > 0, c > 0, \text{则}$$
$$\lim_{n\to\infty} \sqrt[n]{a^n + b^n + c^n} = \max\{a, b, c\}$$

$$x > 0$$

$$\text{求} \lim_{n\to\infty} \sqrt[n]{x^n + x^{2n}}.$$

$$\begin{aligned}\lim_{n\to\infty} \sqrt[n]{x^n + x^{2n}} &= \lim_{n\to\infty} \sqrt[n]{x^n + (x^2)^n} \\ &= \max\{x, x^2\} = \begin{cases} x, 0 < x < 1 \\ x^2, x \geq 1 \end{cases}\end{aligned}$$

$$\text{求极限} \lim_{n\to\infty} (\frac{1}{2n^2+1} + \frac{2}{2n^2+2} \ldots + \frac{n}{2n^2+n}).$$

$$b_n = \frac{1}{2n^2+1} + \frac{2}{2n^2+2} \ldots + \frac{n}{2n^2+n}$$

$$\because \frac{i}{2n^2+n} \leq \frac{i}{2n^2+i} \leq \frac{i}{2n^2+1} (1 \leq i \leq n)$$

$$\therefore \frac{1+2+\ldots+n}{2n^2+n} \leq \frac{1}{2n^2+1} + \frac{2}{2n^2+2} + \ldots + \frac{n}{2n^2+n} \leq \frac{1+2+\ldots+n}{2n^2+1}$$

$$\text{即} \frac{1}{2} \frac{n(n+1)}{2n^2+n} \leq b_n \leq \frac{1}{2} \frac{n(n+1)}{2n^2+1}$$

$$\because \lim_{n\to\infty} \text{左} = \frac{1}{2} \lim_{n\to\infty} \frac{1+\frac{1}{n}}{2+\frac{1}{n}} = \frac{1}{4}$$

$$\lim_{n\to\infty} \text{右} = \frac{1}{2} \lim_{n\to\infty} \frac{1+\frac{1}{n}}{2+\frac{1}{n^2}} = \frac{1}{4}$$

$$\therefore \text{原式} = \frac{1}{4}$$

单调有界数列必有极限

$$\{a_n\} \uparrow \begin{cases} \text{无上界} \Rightarrow \lim_{n\to\infty} a_n = +\infty \\ a_n \leq M \Rightarrow \lim_{n\to\infty} a_n \exists \end{cases}$$

$$\{a_n\} \downarrow \begin{cases} \text{无下界} \Rightarrow \lim_{n\to\infty} a_n = -\infty \\ a_n \geq M \Rightarrow \lim_{n\to\infty} a_n \exists \end{cases}$$

$$\text{设} \{a_n\} = \sqrt{2}, a_2 = \sqrt{2+\sqrt{2}}, a_3 = \sqrt{2+\sqrt{2+\sqrt{2}}}, \ldots, \text{证明：数列} \{a_n\} \text{收敛，并求其极限.}$$

1. $a_{n+1} = \sqrt{2+a_n} (n = 1, 2, \ldots)$
2. $\{a_n\} \uparrow$
3. 现证$a_n \leq 2$

   $a_1 = \sqrt{2} \leq 2$, 设$a_k \leq 2$, 则

   $a_{k+1} = \sqrt{2+a_k} \leq \sqrt{2+2} = 2$

   $\therefore \forall n$, 有$a_n \leq 2 \Rightarrow \lim\limits_{n\to\infty} a_n \exists$
4. 令 $\lim\limits_{n\to\infty} a_n = A$

   $a_{n+1} = \sqrt{2+a_n} \Rightarrow A = \sqrt{2+A}$

   $\Rightarrow A^2 - A - 2 = 0 \Rightarrow A = -1(\text{舍}), A = 2$

$$a_1 = 2, a_{n+1} = \frac{1}{2}(a_n + \frac{1}{a_n}), \text{证：} \lim_{n\to\infty} a_n \exists$$

1. $\because a_n + \dfrac{1}{a_n} \geq 2$

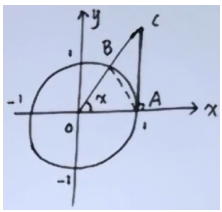
$\therefore a_{n+1} \geq 1$

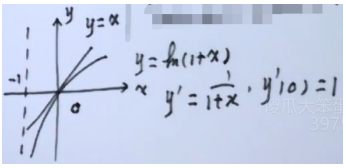
而$a_1 = 2 \geq 1, \therefore a_n \geq 1$

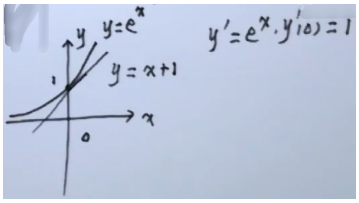
2. $a_{n+1} - a_n = \dfrac{1}{2}(a_n + \dfrac{1}{a_n}) - a_n = \dfrac{1 - a_n^2}{2a_n} \leq 0$

$\Rightarrow \{a_n\} \downarrow \Rightarrow \lim\limits_{n\to\infty} a_n \exists$

重要极限







1. $0 < x < \dfrac{\pi}{2}$时, $0 < \sin x < x < \tan x$

$S_{\Delta AOB} = \dfrac{1}{2}\sin x$

$S_{扇AOB} = \dfrac{1}{2}x$

$S_{Rt\Delta AOC} = \dfrac{1}{2}\tan x$

2. $x > -1$时, $\ln(1+x) \leq x$
3. $x \in (-\infty, +\infty), e^x \geq 1 + x$

$$\lim_{\Delta\to 0} = \frac{\sin\Delta}{\Delta} = 1$$

$$\lim_{\Delta\to 0}(1+\Delta)^{\frac{1}{\Delta}} = e$$

$$\lim_{x\to 0}\frac{1-\cos^3 x}{x\ln(1+2x)}$$

$$\begin{aligned}
原式 &= \lim_{x\to 0}\frac{1-\cos^2 x}{2x^2} \\
&= \frac{1}{2}\lim_{x\to 0}(1+\cos x)\frac{1-\cos x}{x^2} \\
&= \frac{1}{2} * 2 * \frac{1}{2} \\
&= \frac{1}{2}
\end{aligned}$$

$$\lim_{x\to 0}\frac{e^{\tan x}-e^{\sin x}}{x\arcsin^2 x}$$

$$\begin{aligned}
\text{原式}&=\lim_{x\to 0}\frac{e^{\tan x}-e^{\sin x}}{x^3}\\
&=\lim_{x\to 0}e^{\sin x}\frac{e^{\tan x-\sin x}-1}{x^3}\\
&=1*\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}\\
&=\lim_{x\to 0}\frac{\tan x}{x}*\frac{1-\cos x}{x^2}\\
&=\frac{1}{2}
\end{aligned}$$

$$\lim_{x\to 0}\frac{\sqrt{1+x\cos x}-\sqrt{1+x}}{x^3}$$

$$\begin{aligned}
\text{原式}&=\lim_{x\to 0}\frac{1}{\sqrt{1+x\cos x}+\sqrt{1+x}}\frac{x\cos x-x}{x^3}\\
&=\frac{1}{2}\lim_{x\to 0}\frac{\cos x-1}{x^2}\\
&=-\frac{1}{4}
\end{aligned}$$

$$\lim_{x\to 0}(1-\sin 2x^2)^{\frac{1}{x^2}}$$

$$\begin{aligned}
\text{原式}&=\lim_{x\to 0}[(1-\sin 2x^2)^{-\frac{1}{\sin 2x^2}}]^{\frac{1}{x^2}*(-\sin 2x^2)}\\
&=e^{-\lim_{x\to 0}\frac{\sin 2x^2}{x^2}}\\
&=e^{-2}
\end{aligned}$$

$$\lim_{x\to 0}(\cos x)^{\frac{1}{x\ln(1-x)}}$$

$$\begin{aligned}
\text{原式}&=\lim_{x\to 0}\{[1+(\cos x-1)]^{\frac{1}{\cos x-1}}\}^{\frac{\cos x-1}{x\ln(1-x)}}\\
&=e^{\lim_{x\to 0}\frac{\cos x-1}{x\ln(1-x)}}\\
&=e^{\lim_{x\to 0}\frac{-\frac{1}{2}x^2}{-x^2}}\\
&=e^{\frac{1}{2}}
\end{aligned}$$

$$\lim_{x\to 0}(\frac{1+\tan x}{1+\sin x})^{\frac{1}{x^3}}$$

$$\begin{aligned}
\text{原式}&=\lim_{x\to 0}\{[1+(\frac{1+\tan x}{1+\sin x}-1)]^{\frac{1}{\frac{1+\tan x}{1+\sin x}-1}}\}^{\frac{1}{x^3}(\frac{1+\tan x}{1+\sin x}-1)}\\
&=e^{\lim_{x\to 0}\frac{1}{x^3}(\frac{\tan x-\sin x}{1+\sin x})}\\
&=e^{\lim_{x\to 0}\frac{1}{1+\sin x}\frac{\tan x-\sin x}{x^3}}\\
&=e^{\lim_{x\to 0}\frac{\tan x}{x}\frac{1-\cos x}{x^2}}\\
&=e^{\frac{1}{2}}
\end{aligned}$$

$$\lim_{x\to\infty}(\frac{2x^2+3x+1}{x-1}-2x)$$

$$\begin{aligned}
\text{原式} &= \lim_{x\to\infty}\frac{2x^2+3x+1-2x(x-1)}{x-1} \\
&= \lim_{x\to\infty}\frac{5x+1}{x-1} \\
&= \lim_{x\to\infty}\frac{5+\frac{1}{x}}{1-\frac{1}{x}} \\
&=5
\end{aligned}$$

$$\lim_{x\to\infty}(\sqrt{x^2+4x+1}-x)$$

$$\begin{aligned}
\text{原式} &= \lim_{x\to\infty}\frac{4x+1}{\sqrt{x^2+4x+1}+x} \\
&= \lim_{x\to\infty}\frac{4+\frac{1}{x}}{\sqrt{1+\frac{4}{x}+\frac{1}{x^2}}+1} \\
&=2
\end{aligned}$$